CS 1XD3 Lab 2.1: Your First Web Pages

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In this lab, you will get started on your **web portfolio** using basic **HTML** and **CSS** from the first week's lectures. Then you will log on to the **course web server** and use **FileZilla** to host your files.

Before You Start

Download and install **Visual Studio Code** and the **FileZilla Client**. Then get **template.html** from the **Full Stack Example Pack**. The links to all three are on Avenue under Course Resources.

computerhope.com

A. Hello, World!

In this tutorial, you will create a simple HTML file and then host it on the course web site.

1. Create an index.html File

a. Copy **template.html** to a new file called **index.html**, then edit with **Visual Studio Code** to change the **title** and **p** element so that they both have the contents "Hello, World!".
b. Watch for **syntax errors** in **Visual Studio Code**.
c. **Open** by double-clicking **index.html** in your file explorer, or by choosing "Run Without Debugging" from the **Visual Studio Code** "Run" menu.

2. Get McMaster Virtual Private Networking

If you are connected to the McMaster Wi-Fi, you can skip this step. If you connect from home, you need to use the McMaster VPN. Instructions are here.

3. Connect to the cs1xd3 Server

This is a one-time step to set up your account on the course web server.

a. In windows, open the **Command Prompt** (**cmd**). In MacOS, open the **Terminal**.
b. Type **ssh *your_macid*@cs1xd3.cas.mcmaster.ca** at the prompt.
c. It can take a minute or two to connect. If it times out, it could be because you're not connected to McMaster Wi-Fi and you don't have the VPN running. See step 2.
d. If prompted about security certificates, answer "yes".
e. When prompted, enter your macid password. **YOU WILL GET NO FEEDBACK**. That's expected. Just keep typing and hit **enter**.
f. If you know **Linux** feel free to look around, otherwise just close the window.

> **Gotcha**
>
> The cs1xd3 server is running **Linux**. Linux is not very communicative. You will get no feedback when typing passwords. Just keep typing, it is listening to you.

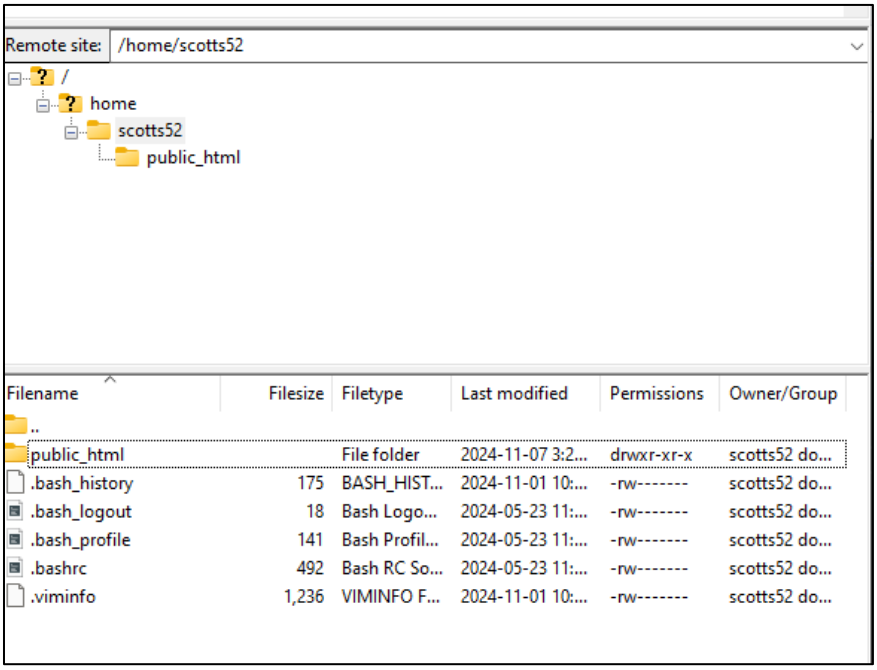
4. Connect to cs1xd3 with FileZilla

FileZilla is an **FTP Client**. **FTP** stands for "**File Transfer Protocol**". It is a standard communication protocol for transferring files from one machine to another. We will use the **Secure** version of the **FTP** protocol, also known as **SFTP**. This protocol encrypts files during transfer.

a. Run FileZilla and fill in the information in the top bar...
   - **Host:** cs1xd3.cas.mcmaster.ca
   - **Username and Password:** Your MacID username and password
   - **Port:** 22 (this configures FileZilla to use SFTP)

b. Press the **Quickconnect** button (next time, cs1xd3 will be in your **Quickconnect history**)
   - If it times out, it could be because you're not connected to McMaster Wi-Fi and you don't have the VPN running. See **Step 2** above.
   - If prompted about security certificates, answer "yes". It's ok.

c. Once connected, you are looking at your computer's **local** file system on the left and the **cs1xd3 remote** file system on the right. The **right side** might look something like this:



d. The **public_html** folder (**directory**) is your public web folder. Anything you put in there is publicly accessible on the web by default.

e. The **".."** folder is the parent of the current folder.

5. Host Your index.html File

a. Navigate to your **public_html** folder in **FileZilla**.
b. If you see any files there, delete them.
c. Drag **index.html** from your local machine into the **public_html** folder. You can drag it from the left-hand-side **local** view, or you can drag it from anywhere else on your computer.
d. That's it – you just hosted your web site.

6. Visit index.html on the Web

a. In a browser, go to **https://cs1xd3.cas.mcmaster.ca/~your_macid**, inserting your own macid where required.
   - This is a **public URL** that points to your **public_html** folder.
   - You should see "hello, world" displayed in the browser. If you get an error, see the troubleshooting information below.
   - If you **right click** the page and choose "**view page source**", you will see the original HTML code of your **index.html** file.

b. In the above, you told the server which **folder** you wanted, but not which **file**. By default, if you request a folder, the web server will search in it for a file called "index".
   - Try this alternate URL: **https://cs1xd3.cas.mcmaster.ca/~your_macid/index.html**
   - Go back to **FileZilla** and rename **index.html** to **hello.html** and reload **https://cs1xd3.cas.mcmaster.ca/~your_macid**. What do you see now? Does this make sense?

c. Rename the file to **index.html** again.

7. Troubleshooting: File Permissions

If the browser gives you a "**forbidden**" or "**permission denied**" error, it's because the file or the folder do not have the correct **public permissions**.

- Right click **index.html** in **FileZilla** and select "**File Permissions**". Make sure the file has public **read** permission turned on. Then reload the page.
- If it's still not working, check the **public_html** folder's permissions. It should have public **read** and **execute** permissions turned on. You could also check your home folder, which should have public **read** permission.

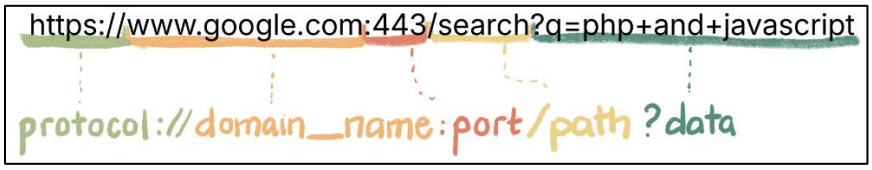
Bugs in the Instructions

If your experience doesn't match this guide, or if you find a problem that's not covered here, please let us know so we can update the instructions.

B. Paths and URLs

In this tutorial you will explore the relationship between your local file structure and the public URL of your web site.

If you haven't already, this would be a good time to review Chapter 4 of Full Stack Web Development for Beginners. The basic parts of a URL are labelled below. See the textbook for more explanation.



a. In **FileZilla**, create a new **folder** (**directory**) called "test" inside **public_html**. You can do this by right-clicking inside the **public_html** folder and choosing "Create Directory".
   - On the **cs1xd3** server, this folder is **/home/your_macid/public_html/test**. That's the **absolute path** to your new folder on **cs1xd3**
   - On the web, this folder is **https://cs1xd3.cas.mcmaster.ca/~your_macid/test**. That's the **public URL** of your new folder
   - Note the mapping (in blue) between the first part of the URL and your **public_html** folder.

b. Create a new **index.html** file with the contents "Hello, Test" and place it in the **test** folder.
   - When you visit the **test** folder through a URL, it will automatically load **index.html**

c. **A general principle:** every subfolder on your web site should have its own **index** file.

C. Password Protecting Your Work

In this tutorial you will password protect a web folder. By default, web sites are public (although our server is restricted to the McMaster community) so, password protection is a good idea for any substantial work you are handing in for credit. If you don't password protect it, other people might be able to find and copy your work. You might also want to password protect if you are uncomfortable with anyone other than you and the graders viewing your work.

1. Create a UserID and Password

First, you will create a file of **userids** and **passwords**.

a. From a **Windows command prompt** or **Mac terminal**, log in to the **cs1xd3** server:

```
ssh your_macid@cs1xd3.cas.mcmaster.ca
```

b. To add the first **UserID** and **Password**, execute the following command:

```
htpasswd -c ~/.htpasswd grader
```

This creates a new password file called **.htpasswd** in your home folder (**~**) and creates the "**grader**" UserID. It will ask you to type a password for **grader**.

c. If you want to add more UserIDs and passwords, use the same command but without the -c flag (this flag causes a new file to be created).

```
htpasswd ~/.htpasswd new_userid
```

2. Viewing the .htpasswd File (optional)

a. The **ls** command lists files and folders in the current folder, but it won't show **.htpasswd** because filenames that start with a dot are hidden in Linux. Use the **ls -a** command instead.

b. If you want to see the contents of **.htpasswd**, use the command **cat .htpasswd**. You will see one line per **userid** with a **hashed** version of each password, something like this:

```
user:$apr1$DPmnpFx6$OAZJJe20t0OPKm3RMRlxp/
user2:$apr1$eNKWx75D$dFDjK3YIu6X/0PRGbaWvW0
```

The passwords are stored like this to make them hard to crack. Hashing is a form of one-way encryption in which converting from a password to the hashed version is easy, but it is very difficult to convert a hashed password back to its original form.

3. Password Protecting a Web Folder

Now you will tell a web folder to use your file of userids and passwords for authentication.

a. Get the **htaccess** file from Avenue and change the text so that it uses your macid. Here's the text with the part you have to change in blue:

```
AuthName "Dialog prompt"
AuthType Basic
AuthUserFile /home/your_macid/.htpasswd
Require valid-user
```

b. From FileZilla (or from the Linux command line if you are comfortable with it), go to your **public_html/test** folder and upload the **htaccess** file, then rename it to **.htaccess** (the "." makes it a hidden file).

c. Try it out at this URL: **https://cs1xd3.cas.mcmaster.ca/~your_macid/test**. You should get a password prompt. Use one of the **userids** you created in step 1.

d. You can add the same **.htaccess** file to any folder within your web site that you want to protect. If you want to protect the entire web site, put **.htaccess** in the **public_html** folder.

4. Advanced: Managing Passwords

a. To change or add a password:

```
htpasswd ~/.htpasswd userid
```

b. To set up different passwords for different folders:
   - Redo Step 1, but use a different file name, such as **.htpasswd2**.
   - In each protected folder, change the **AuthUserFile** line in **.htaccess** so that it points to the password file you want to use for that folder.

D. Your Portfolio

A hosted portfolio page is your deliverable for this lab. If you run out of time during the lab period, you should finish this on your own time.

It's time to make something better than a hello world page. We haven't introduced a lot of HTML and CSS yet, but you have enough to begin creating a portfolio web page. This web page should be the "front door" to your web site. Your portfolio should be the **index.html** file inside your **public_html** folder. It will contain information about you and links to other local and remote files and folders showcasing your work.

1. HTML for Structure

Follow the instructions in the **Exercises** section of **Chapter 6** of the textbook. Ignore the part about XAMPP, we're not using that yet.

In exercise 1g, where it says to include at least one link to another **HTML** file in the portfolio folder, you should create two new pages and link them to your index.html page:

- A **work.html** page that will have links to your course work. For now, it can be mostly empty or say "coming soon".
- A **journal.html** page that will have journal entries where you reflect on your progress. The first journal entry should have today's date and should be a short statement about your prior web experience and your feelings about the course so far.

It's up to you how you create the pages, but if you copy **index.html** and then modify it, you'll have a better chance at making the three pages look like they belong together. Think about using a common header, footer, and links (navigation) section on all three pages...

2. CSS for Style

Follow the instructions in the **Exercises** box in **Chapter 7** of the textbook. Use an **external stylesheet** and import it on all three pages of your portfolio to give them all a similar look and feel.

We haven't really covered any CSS for layout yet, so your web pages will probably just look like colorful word documents rather than modern and professional web pages. That's ok, just do the best you can with the tools you have. Next week you'll be able to make it look much better.

3. Host your Portfolio Page

Transfer all the files (**index.html**, **work.html**, **journal.html**, **images**, and **css** files) to your **public_html** folder on **cs1xd3**. You should have your CSS files in a **css** folder and images in an **images** folder. You can password protect your site if you want to keep it private, or you can leave password protection off to show the whole (McMaster) world.

4. Test your Page

You're not done until you have visited all your pages through the public URL on a browser. Sometimes there are problems with the page that you don't notice until you have uploaded it (for example, Linux is case sensitive while Windows is not, so images and links that work locally might not work on Linux). Test every page and subpage to make sure it works!

IMPORTANT: Host Your Portfolio Correctly!

If https://cs1xd3.cas.mcmaster.ca/~your_macid does not automatically load your portfolio, then you have not hosted it correctly. See the previous tutorials or get help from the TA, and try again!

5. Hand in the URL

Hand in the URL to your portfolio using the **Portfolio Web Site** assignment on Avenue. If you password protected it, include the **userid** and **password** the graders will need in order to access your work.